Shivam Agarwal

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Education

Chandigarh University, Punjab — **2023 – 2027**
Bachelor of Engineering in Computer Science and Engineering — **CGPA: 8.11 / 10.0**

Vidya Mandir Sr. Sec. School, Rajasthan — **2020 – 2021**
Intermediate (CBSE Class XII) — **Percentage: 80.6%**

Technical Skills

Languages: Java, SQL, JavaScript (ES6+), Python, C++

Backend Frameworks: Spring Boot, Spring MVC, Spring Security (JWT), Spring Data JPA, Hibernate, RESTful APIs

Frontend & Web: React.js, HTML5, CSS3, Responsive Web Design

Databases: MySQL, PostgreSQL

Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, OS, Networks

Developer Tools: Git, GitHub, Maven, Postman, IntelliJ IDEA, VS Code

Projects

AI Resume Analyzer | *Spring Boot, Google Gemini API, Spring Data JPA, MySQL* [GitHub]

- Engineered an automated ATS resume scoring service using **Spring Boot**, exposing **8+ RESTful endpoints** for multi-format document evaluation against job descriptions.
- Integrated **Google Gemini 1.5 API** using structured prompt engineering to parse resumes, identify keyword gaps, and return JSON-formatted scoring analytics.
- Implemented document extraction with **Apache PDFBox** and designed normalized relational schemas using **Spring Data JPA** to persist evaluation history and reduce redundant LLM calls.

Smart Seat & Ticket Reservation Platform | *Spring Boot, React.js, MySQL, Spring Security* [Live Demo]

- Developed a full-stack seat booking web application featuring interactive seat availability layouts and automated ticket confirmations.
- Engineered transactional REST APIs in **Spring Boot** with **concurrency control** (@Transactional) to prevent race conditions and eliminate double-booking.
- Built an interactive client dashboard with **React.js** and secured endpoints using **Spring Security with JWT** for role-based access control (RBAC).

LRU Cache & Algorithm Visualizer | *React.js, JavaScript, Data Structures, CSS* [Live Demo]

- Implemented custom **HashMap and Doubly Linked List** data structures from scratch to achieve verified $\mathcal{O}(1)$ time complexity for `get()` and `put()` operations.
- Visualized cache hits, misses, and evictions using dynamic step animations and integrated real-time hit-ratio analytics to benchmark cache efficiency.

Achievements & Coding Profile

- **Data Structures & Algorithms:** Solved **250+ problems** across LeetCode and Codeforces.
- **Academic Consistency:** Maintained a strong academic record with an **8.11 CGPA** in Computer Science Engineering at Chandigarh University.